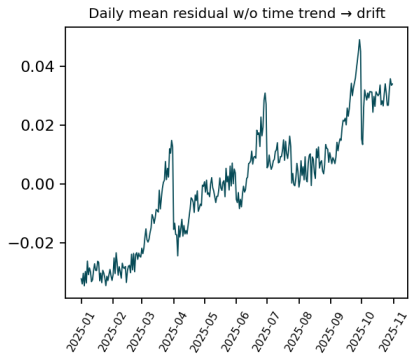
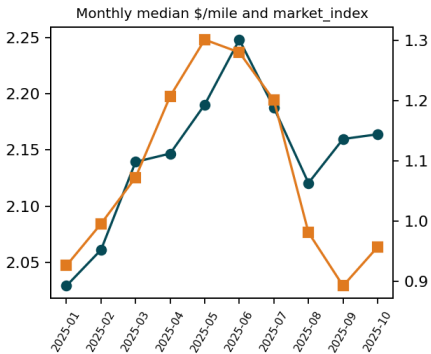
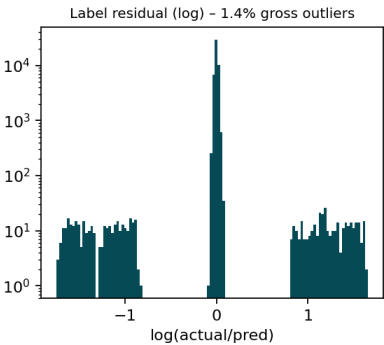


Freight Rate Prediction – Approach & Validation Report

Candidate: Stuti · Role: Machine Learning Engineer · Code: see repository (README.md → python train.py)

1. Data and key findings

- **Labeled data:** 48,000 loads, 2025-01-01 → 2025-10-31, 64 cities. **Validation set:** 12,000 loads, 2025-11-01 → 2025-12-31, i.e. strictly *after* the labeled period.
- Rate is essentially **distance × a per-mile rate** (corr 0.91 with distance); the per-mile rate depends on equipment (Dry Van < Flatbed < Reefer), short-haul premium, lane/city geography, and a market level.
- **market_index** is a day-level signal measured with row-level noise (sd≈0.025 around the daily mean). The daily means explain most of the day-to-day rate movement; **quote_signal** adds a smaller, U-shaped effect (rates are higher when it is far from ≈2.0).
- After removing market effects, rates still **drift upward** ≈ **+8% over the year** (about 0.022% per day). This matters because validation lies beyond the training date range.
- Noise floor: for ≈98.6% of rows, the best model's log-error sd is ≈2%; the remaining ≈1.4% of labels are gross outliers (rate ×0.2–0.4 or ×2.3–5, unrelated to any feature).



2. Data-quality issues and handling

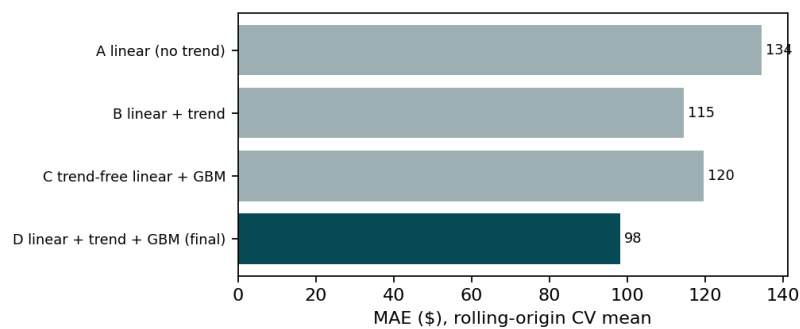
Issue	Evidence	Handling
Negative weights	292 train / 145 val rows; magnitudes look like normal weights	Use weight (sign error); keep a flag feature
Missing weight	300 train / 165 val (0.6–1.4%)	Median impute + missing flag
Missing market_index	374 train / 249 val	Impute with same-day mean of other loads (features only) + flag
Weight capped at 47,500	1,204 train rows exactly at the cap	Left as is (tree/linear handle it)
Gross label outliers	≈1.4% of rows: ×0.2–0.4 or ×2.3–5 off the model	Excluded from fitting (robust Huber pass flags resid >0.4 in log space); L1 loss in GBM
Unseen cities in validation	8 cities (≈12% of val rows) never appear in training	No city-ID features; lat/lon + distance only. Verified with held-out-city simulation
Distance vs coordinates	Coordinates are jittered/synthetic: haversine ≠ distance; 70-mile floor	Trust `distance` (consistent per lane, ±2%); coordinates only used as geography features
No duplicate IDs / duplicate rows	checked	none needed

3. Train / validation split strategy

The final validation set is **later in time** (Nov–Dec) than every labeled row, so a random K-fold split would be optimistic: it would let the model see the same days/market conditions it is tested on. I therefore used **forward-chaining (rolling-origin) cross-validation**: train on all data before a cutoff and predict the next two months (the same horizon as the real task), for cutoffs 1 May, 1 Jul and 1 Sep. Metrics are averaged across the three folds. Because 12% of validation rows involve cities absent from training, I also ran a **held-out-city check**: train without 8 random cities, test on Sep–Oct. Error on rows touching unseen cities (MAE $\approx$ \$98, MdAPE 1.57%) was essentially the same as on seen cities (MAE $\approx$ \$99, MdAPE 1.51%).

Raw labels (including outliers) are used for evaluation because the real validation labels will contain such outliers too; I also report a trimmed MAPE (dropping the worst 1.5% of rows) to show model quality on 'normal' loads.

Model (rolling-origin CV mean)	MAE \$	RMSE \$	MAPE %	MdAPE %	trimmed MAPE %
A linear (no trend)	134.5	638	5.85	3.30	3.56
B linear + trend	114.6	630	5.09	2.35	2.73
C trend-free linear + GBM	119.6	633	5.15	2.62	2.85
D linear + trend + GBM (final)	98.3	626	4.22	1.53	1.85



4. Model and reasoning

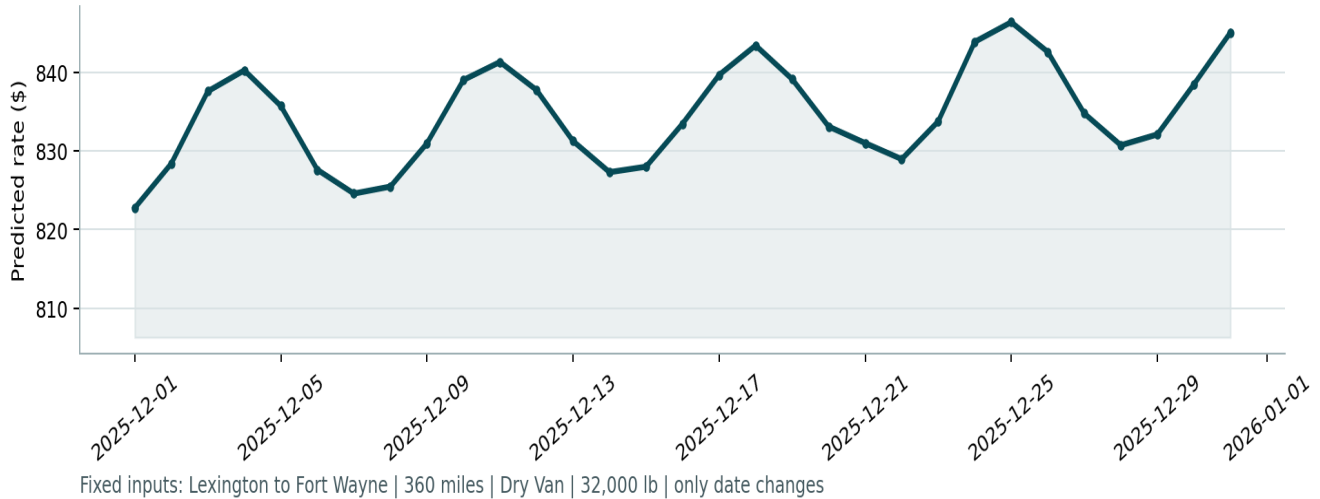
Two-stage model in log-rate space. Stage 1 is a robust (Huber-screened, then Ridge) linear model on log-distance (+quadratic), equipment, weight, market_index, quote_signal terms, day-of-week and an explicit **linear time trend**. This stage carries the market level and drift and, crucially, *extrapolates* into Nov–Dec, which tree models cannot do (they flat-line at the last observed level). Stage 2 is a HistGradientBoosting regressor (L1 loss, 300 iterations, lr 0.05) fit on the stage-1 residuals using geography (lat/lon, deltas), distance, equipment and the other load features; it captures lane/region premiums and non-linear distance effects. No raw date feature is given to the GBM. Predictions are $\exp(\log\text{-rate})$, i.e. a conditional median, which is the right target under the heavy-tailed label noise (median is optimal for MAE/MAPE-type metrics).

Why not a plain GBM? Without the trend stage the validation-like folds show a systematic under-prediction of $\approx 2\%$ (Sep–Oct), and MAE of 110 vs 98 for the final model. Model comparison above: adding the trend cuts MdAPE 3.3 $\rightarrow$ 2.4 (linear), and adding GBM on top 2.4 $\rightarrow$ 1.5, which is already close to the $\approx 1.4\text{--}2\%$ noise floor.

Daily aggregates for the validation/December period (daily mean market_index, quote_signal) are computed only from the *feature* columns of validation.csv – no labels are used. For the December chart (one row per date), market_index and quote_signal for each date are the December daily means from validation.csv; weight, distance, equipment and lane are the fixed inputs.

5. Fixed December prediction chart (produced by the provided score.py)

Candidate: December 2025 Predicted Load Rate



Predicted Lexington → Fort Wayne (Dry Van, 360 mi, 32,000 lb) rate ranges ≈ \$823–\$846 with a weekly cycle (mid-week higher, weekend lower) and slightly rising market level; this is in line with this lane's Jan–Oct history (≈ \$780–\$930 monthly medians).

6. Limitations / what I would do next

- The time trend is estimated from only 10 months and extrapolated 2 months; if the drift is really seasonal rather than linear, Nov–Dec error could be ≈1–2% higher. A longer history would let me separate seasonality from trend.
- Outlier rows (≈1.4%) are inherently unpredictable and dominate RMSE; a classifier cannot find them from the features (outlier rate is flat across equipment/distance bins). If the scoring metric is RMSE, expect a large irreducible component.
- Possible extensions: lane-level target encodings with shrinkage for frequently-seen lanes, quantile outputs for prediction intervals, and monitoring for drift in production.